



BATTERY (PCB-001, owner decisions 2026-10-04)

- One protected LiPo pouch cell, 1000 mAh or more, rated 0.5 C charge charging 0 to 45 C, JST-PH plug, + on pin 1 = RED WIRE, pin 3 = cell thermistor (remove R12 if used). Charge current 494 mA (R14 1.0k 3.6k for 500 mAh, up to 274 mA at tolerance).
- NO ON-BOARD REVERSE-POLARITY GUARD (rev 0.6, as on the Pico W), F1 polyfuse in the cell lead (rev 0.7) contains shorts and limits heat after a wiring mistake but does NOT save the two chips a reversed cell destroys. The keypad plug, the silkscreen, one approved cell listing and checking the wire colours BEFORE plugging in are the protection: a mis-wired cell can damage the DW04k and the charger's BAT pin (-0.3 V limits), UNPLUG USB BEFORE SWAPPING CELLS (the socket's 5V line from the charger).
- The charger does not sense the cell's temperature with a two-wire cell (R12 10k fitted = check bypassed): the cell's protection board and the 0.5 C rate protect it, safely timer about 5 h, longer while the charger throttles (DPM, thermal).
- First power-ups: current-limited supply, no cell; then cell alone with the meter; then both. Charge on a non-flammable surface, never unattended. A tripped F1 resets only after the cell is unplugged.